

Introduction to Software Engineering

Project Proposal

The student team is required to complete the **Project Proposal** documentation for the assigned course project, following the attached template.



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Project Proposal

Objectives

This document focus on the following topics:

- ✓ Completing the Project Proposal document with the following sections:
 - Preliminary Problem Statement
 - Proposed Solution
 - Development Plan
 - Human Resources & Costing Plan
- ✓ Understanding the Project Proposal document.

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Member Contribution Assessment

ID	Name	Contribution (%)	Signature
23127150	Lê Quốc Anh	100%	
23127205	Lâm Hữu Khánh	100%	
23127215	Trần Tuấn Kiệt	100%	
23127326	Lê Mai Hoài Bảo	100%	
23127407	Trần Hữu Lộc	100%	

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Preliminary Problem Statement

Sợi Chỉ City is an online social networking platform that seamlessly combines **short video** and **image sharing**, designed to provide users with a **creative, expressive**, and **interactive digital space**. In an era where most social media platforms focus on a single form of content, **Sợi Chỉ City** was developed to overcome this limitation by allowing users to share two main types of content simultaneously: **short videos** and **images**. The platform aims to foster a **vibrant community** where users can freely express their opinions, share life moments, spread emotions, and connect with others who share similar interests or perspectives.

When creating content, users can choose to **upload videos or images**, attach **descriptions, hashtags**, and select a visibility mode (public, friends, or private). The system enables users to edit, delete, or pin their posts, as well as **interact** with others through **likes, comments**, and **shares**. Additionally, users can follow other accounts, build friend lists, and organize their favorite content within their personal profiles.

The system's frontend is built using **React, Tailwind CSS**, and **Shadcn**, ensuring a modern, responsive, and **user-friendly interface** compatible with **HTML5-supported web browsers**. The backend is powered by **FastAPI (Python)** running on a **Uvicorn server**, providing high performance and scalability. All user data, content, and interactions are stored in **Firestore (Firebase)**, enabling **real-time access** and seamless synchronization across devices. The platform is primarily developed for the **web environment**, with the potential for future expansion to mobile applications.

With **Sợi Chỉ City**, the development team envisions a **social network** where users not only entertain themselves but also share meaningful values, emotions, and ideas in a concise, authentic, and engaging way. This project represents the first step toward building a creative, inclusive, and expressive online community for the digital age.

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Proposed Solution

3.1 Software

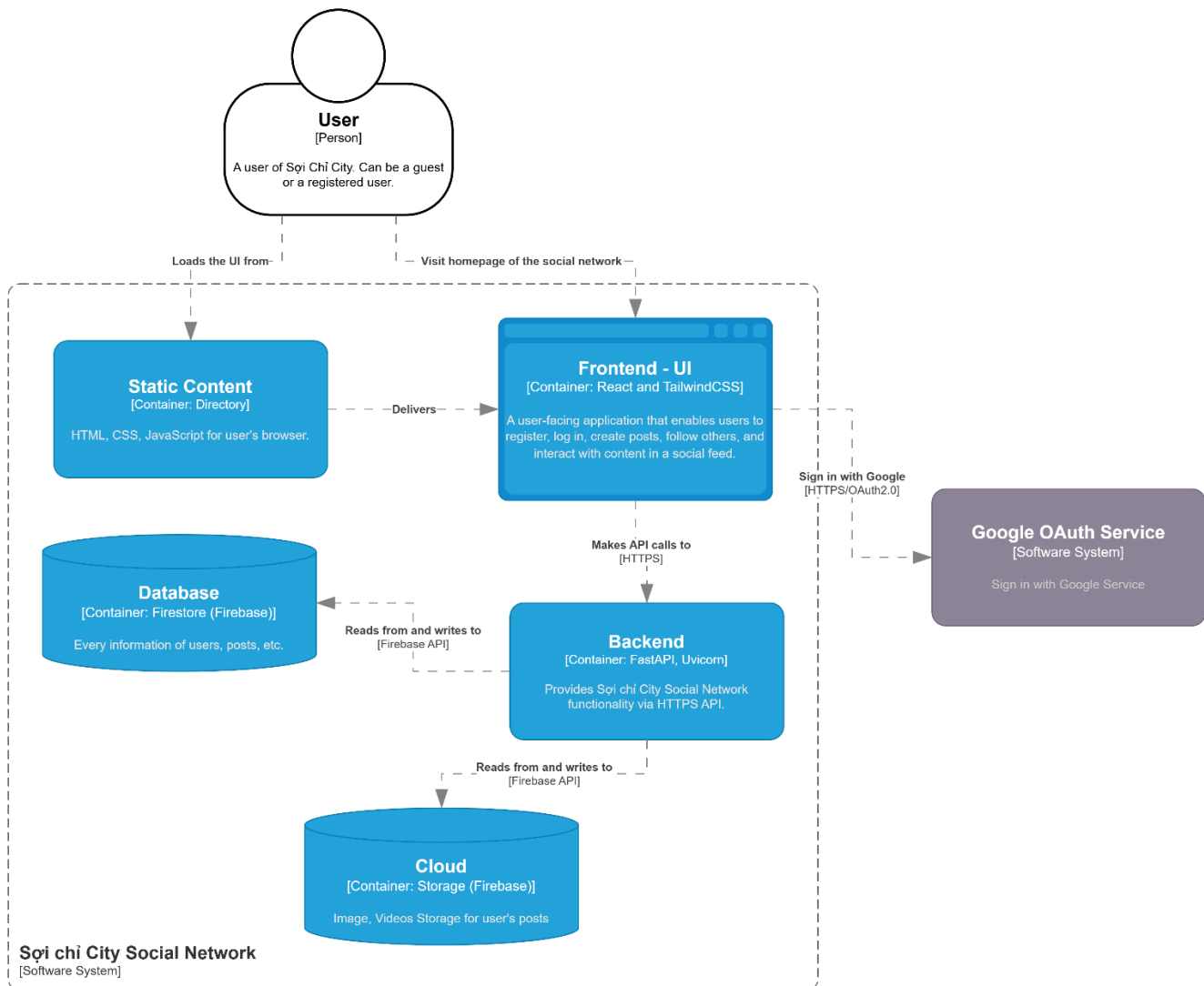
3.1.1. Features

Needs	Requirements
As a new user , I want to sign up for an account using my email and a secure password.	User Registration (Email)
As a new user , I want to quickly register using my existing Google account.	Google Registration
As a registered user , I want to log in to my account using my email and password.	User Login (Credentials)
As a user , I want to securely reset my password if I forget it.	Password Recovery
As a registered user , I want to change my profile picture by uploading a new image or edit my profile information.	Edit User Profile
As a user , I want to follow another user to see their posts on my "Following" feed.	Follow User
As a user , I want to unfollow an account that I no longer wish to see content from.	Unfollow User
As a user , I want to view posts when I access to homepage and scrolling to explore more posts.	View Posts

As a user, I want to publish a new post containing text, images and short videos.	Post Creation
As a user, I want to react to a post (e.g., Love, Hate) to express my feelings.	Post Reaction System
As a user, I want to write a text comment on a post to share my thoughts.	Create Comment
As a user, I want to reply directly to another person's comment to create a nested discussion.	Threaded Replies
As a user, I want to "re-post" another user's content to my own profile/feed.	Repost (Internal Share)
As a user, I want to share a link to a post on an external application (e.g., Messenger, Zalo).	External Share
As a user, I want to save a public image or video from a post directly to my device's gallery.	Content Download
As a user, I want to search for other users by their username or display name.	User Search
As a user, I want to search for a hashtag (e.g., #hcmus) or a keyword (e.g., "Nhập môn SE") to discover relevant posts and users.	Global Keyword/Hashtag Search

3.1.2. Software Architecture

Sơ đồ chỉ City's overall architecture using Container-C4 diagram:



3.2 Hardware

User/Client side	Software/Server side
- Device: Desktop, laptop, tablet or smartphone. - Network: Wifi, 4G, 5G with good internet connection.	- Standard server with CPU and RAM are sufficient to handle multiple users. - Enough storage for application data.

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Development Plan

4.1 Requirements Analysis

Objective:

- Clarify the project's objectives and scope (MVP - Minimum Viable Product) with all **stakeholders**.
- Collect, categorize, and analyze functional requirements (posting videos/images, viewing feed, interacting) and non-functional requirements (responsiveness, security, video loading performance).
- Clearly define the core features and assign roles and tasks to each team member.

Deliverables:

- **Requirements Specification Document (RSD)**: Includes User Requirements, Use Case Diagrams for main features, and non-functional constraints.
- **UI Sketches**: Low-fidelity sketches/dummies in **Figma** for core feature flows (Feed, Post, Post Details).
- **Project Management Plan**: Repository structure on **GitHub**, task board setup on **JIRA**.

4.2 Software Design

Objectives:

- Build a consistent, modern, and user-friendly UI design using **Figma**, adhering to the **Shadcn/Tailwind CSS** design system.
- Design an efficient 3-tier system architecture, clearly separating the Frontend (**React**), Backend (**FastAPI**), and Data (**Firestore**).
- Design the NoSQL database structure (**Firestore**) and define API endpoints (**FastAPI**) to support business logic.

Deliverables:

- **High-Fidelity UI/UX Mockup:** The complete, approved **Figma** design for all application screens (using Shadcn components).
- **Software Architecture Document (SAD):**
 - **Data Diagram** (Schema) for **Firestore** collections (e.g., users, posts, comments).
 - **API Specification** for **FastAPI** endpoints (e.g., POST /posts, GET /feed).
 - **Sequence Diagram** for complex flows like "Video Upload".

4.3 Implementation

Objective:

- Code the UI components using **React**, **Tailwind CSS**, and **Shadcn**.
- Develop the API endpoints and business logic on the Backend using **FastAPI** (**Python**), running on **Uvicorn**.
- Fully integrate the Frontend, Backend, and Database (**Firestore** with Firebase Storage) into a functional system.

Deliverables:

- **Frontend Source Code (React):** Functional components, such as VideoFeed, PostCard, and UploadModal.
- **Backend Source Code (FastAPI):** Programmed and tested API endpoints (tested via **Postman**).
- **Working MVP:** A version of the system that can run locally and meets the objectives from Phase 1.

4.4 Testing

Objective:

- Conduct thorough testing (Unit Test, Integration Test) to ensure the system is stable, bug-free, and secure.
- Validate that the implemented features meet the requirements specification and improve user experience based on test results.
- Present the product to **stakeholders** to gather feedback and make revisions.

Deliverables:

- **A Tested and Refined Product.**
- **Test Case Suite:** A document of test cases (such as "upload video > 60s", "post empty comment", etc.).
- **Testing Report:** A detailed log of bugs found and fixed, submitted with the final product.

4.5 *Deployment and Maintenance*

Objectives:

- Deploy the final product onto **self-hosted infrastructure**, ensuring full functionality in the production environment.
- Configure environment variables, domains (if any), and ensure cross-device/browser compatibility.
- Collect feedback from end-users (such as friends, instructor) through various channels (such as interviews, forms).
- Analyze feedback to identify areas for improvement and propose potential advanced features for the future.

Deliverables:

- **Final Released Product** with a public access link.
- **User Feedback Forms** and evaluation summaries.
- **A detailed Upgrade and Maintenance Plan.**

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Human Resources & Costing Plan

5.1 Human Resources

Role	Team member	Responsibilities
Project manager	Lê Quốc Anh	Plan and finalize the project concept. Manage and coordinate team activities. Maintain communication with the project supervisor.
DevOps engineer	Lê Quốc Anh Trần Tuấn Kiệt	Responsible for setting up deployment environments, and ensuring smooth, stable product delivery.
Business analyst	Lê Quốc Anh Lâm Hữu Khánh	Acts as the bridge between the development team and other stakeholders.
UX/UI designers	Lê Mai Hoài Bảo Trần Hữu Lộc	Design the product's UX/UI.
Front-end developers	Trần Tuấn Kiệt Lâm Hữu Khánh Lê Mai Hoài Bảo	Responsible for the front-end development of the project.
Database designers	Lâm Hữu Khánh Trần Hữu Lộc	Design the database structure to be used throughout the project.
Back-end developers	Lê Quốc Anh Lê Mai Hoài Bảo Trần Hữu Lộc	Responsible for the back-end development and ensure proper integration between the back-

		end, front-end, and database of the project.
Project testers	Lê Quốc Anh Trần Tuấn Kiệt Trần Hữu Lộc	Test and evaluate the product, including assessing the feasibility of implemented features.
Documentation writer	Lê Quốc Anh Lâm Hữu Khánh Trần Tuấn Kiệt Lê Mai Hoài Bảo Trần Hữu Lộc	Complete the proposal, meeting minutes, and other group reports.

5.2 Costing Plan

Category	Details	Estimated Cost (VND)
Human Resources	Team effort for analysis, design, development, testing, documentation.	0 (student-led project)
Software & Tools	Development tools: Github, Figma, Jira,...	0 (using free/student versions)
Frontend & Backend	- Frontend: React, Tailwindcss, Shadcn - Backend (if applicable): Uvicorn, FastAPI (Python)	0 (using free versions)
Database & Cloud	Using Firestore and Cloud Storage from Firebase	0 (using free versions)
Domain name & Server	- Domain name: namecheap.com - Server: ZHost (VPS).	400,000 – 700,000

Documentation & Printing	Printing proposal, final report, evaluation forms (if required)	50,000 - 100,000
Miscellaneous Expenses	Internet, communication, unexpected minor expenses.	100,000

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Tools setup

One of the main goals of this project assignment is to help you practice teamwork for professional software development. We'll assess and evaluate your effectiveness in team collaboration. Everyone is required to work closely with others in the group to deliver results, and we'll use tools to monitor and assess each group's teamwork performance.

You are required to use the following tools:

- *Moodle: Used for posting and submitting assignments.*
- *Slack (or a similar tool like MS Teams or Discord or Telegram): Used for discussions and interactions among group members for each project.*
 - *TAs will create a Slack project and add all students to it.*
 - *There will be a general channel for everyone, used for general project discussions.*
 - *Each group will have a channel linked to Trello. Any updates from Trello will be posted in this channel. Group members must use this channel to discuss their project.*
 - *Slack has desktop and mobile applications; please open the app and check it frequently.*
- *JIRA (or a similar tool like Azure Boards): Each group will have one board. Tasks are organized into sprints.*
- *Bitbucket, GitHub, or another source control tool: Used to store source code and documentation. Each group will belong to a team on the repository and have the following folders:*
 - */src: Used to store source code.*
 - */docs: Used to store documentation, which has these subfolders:*
 - *management: Storing planning documents, reports (weekly report, project status report, etc.).*
 - *requirements: Storing all requirements, including vision documents and use cases.*
 - *analysis and design: Storing all analysis and design related documents, including software architecture documents, UML models, and UI designs.*

- *test: Storing all test documents such as test plans, test cases, and test reports.*
- */pa: Including subfolders to store submissions. Each subfolder contains one PA submission.*
- *TAs will help set up these tools during the first few lab sessions.*

Important note: *the effective use of tools to support teamwork is crucial for project success. Thus, using tools effectively is an important criterion for grading. Everyone is required to use the tools chosen by the team regularly.*